



## INFO

-  27/02/2002
-  Porto, Portugal
-  +351 961 903 583
-  kikopiressilva@gmail.com
-  xizesse.github.io
-  /in/franciscopiressilva
-  /Xizesse

## AREAS

Mobile robotics · Marine robotics · Motion planning · Decision making · Obstacle detection and tracking · Control system design

## TOOLS

ROS · ROS 2 · Nav2 · Gazebo · ArduPilot · ArduSub · LiDAR · Codesys · Modbus TCP · OPC-UA · MATLAB · CAD · Linux · Git · LaTeX

## LANGUAGES

C++ · C · Python · Java

## SPOKEN

Portuguese (native) · English (fluent) · German (beginner)

# Francisco Pires da Silva

Robotics Engineer — Mobile & Marine Robotics

I'm a robotics engineer and researcher with an MSc in Electrical and Computer Engineering, currently working on autonomous surface vehicle navigation: obstacle detection, motion planning and avoidance. Drawn to mobile robotics by the complexity of the problems, the creativity the solutions demand, and the range of fields it can transform.

## Experience

### Robotics Researcher

INESC TEC — Centre for Robotics and Autonomous Systems (CRAS) 

Jan 2023 – Present | Porto, Portugal

- Developed and validated the full autonomous navigation stack for an ASV — motion planning, decision making and control system design.
- Dynamic obstacle avoidance for heterogeneous surface and aerial vehicles, with LiDAR-based obstacle detection and tracking.
- Collaborative UAV-ASV mission coordination for offshore inspection.
- 30+ days of field testing, deployments and live demonstrations on the water.

Horizon Europe / H2020: AEROSUB (offshore wind inspection) · TALOS (photovoltaic inspection) · ATLANTIS (Atlantic maritime robotics test platform).

## Education

### PhD Coursework, Electrical and Computer Engineering

Sept 2025 – Aug 2026 | University of Porto (FEUP)  
Curricular year completed.

### Exchange Semester, Electrical and Computer Engineering

Oct 2024 – Mar 2025 | Technical University of Munich (TUM), Germany

### MSc in Electrical and Computer Engineering

Sept 2023 – Jul 2025 | University of Porto (FEUP) | GPA 18/20  
Dissertation: *A Velocity Obstacle Avoidance Algorithm for Heterogeneous Robotic Vehicles* — 19/20.

### BSc in Electrical and Computer Engineering

Sept 2020 – Jul 2023 | University of Porto (FEUP) | GPA 18/20

### MIT Portugal Marine Robotics Summer School

Jul 2026 | Faial, Azores

## Leadership & Outreach

### NEEEC-FEUP

Sept 2020 – Aug 2025 | Electrical and Computer Engineering Students' Branch

- Board Member 2023/24 — headed the technology department, leading a team of 20.
- Logistics coordinator for NEEEL-IT, a robotics competition.
- Ran technical workshops (ROS 2, electronics, Arduino) and high-school outreach.

A little more about me, and the projects I've been involved in:

 [xizesse.github.io](https://xizesse.github.io)